

TrES-1b

R-1595

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_190603 · created 2026-08-02T01:32:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458637.9051 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.018
Transit depth (Rp/Rs)^2	0.93 +/- 0.35 [%]
Orbital Inclination (inc)	89.98 +/- 0.57
Airmass coefficient 1 (a1)	1.1312 +/- 0.0098
Airmass coefficient 2 (a2)	-0.114 +/- 0.045
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	None
Optimal Aperture	4.28
Optimal Annulus	5.35
Transit Duration (day)	0.1002 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.096 ± 0.018 vs 0.1358 (deviation 1.43σ)
Duration fit vs published	0.1002 d vs 0.1042 d (deviation 1.43σ)
Mid-time O–C	3.2 min
Depth SNR	3.5
Residual scatter	0.87 %

Light curve

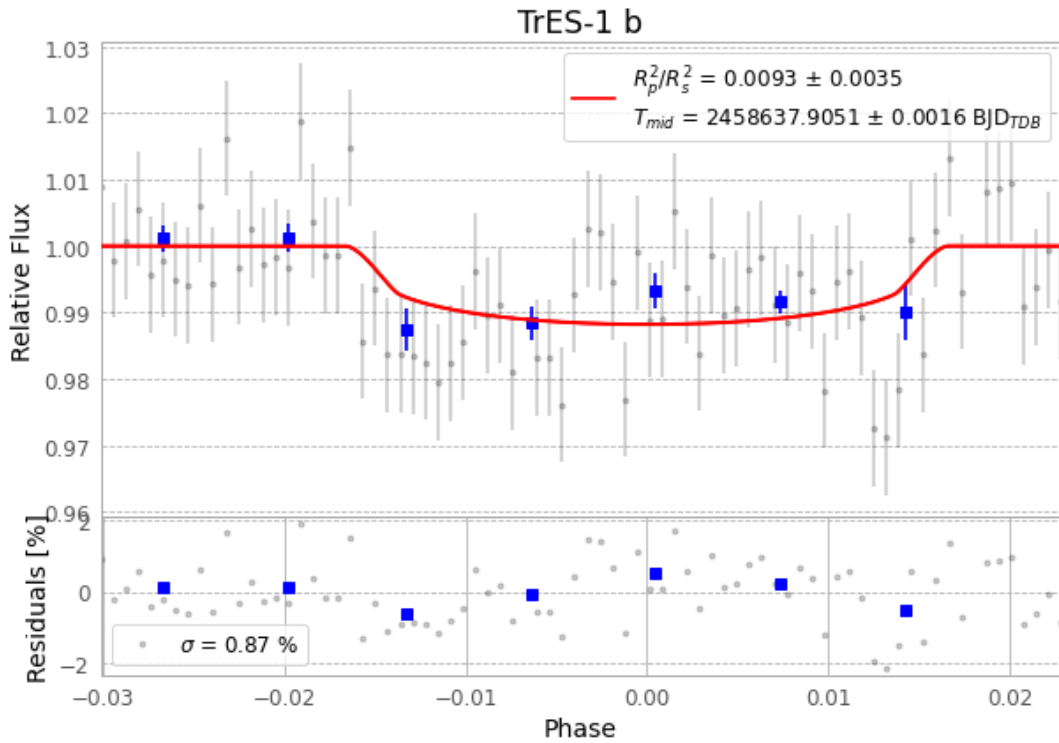


Figure 1 — FinalLightCurve_TrES-1 b_2019-06-03.png (SHA-256 c1016f4827359cf6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1279849d4a4ec47a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

78 FITS frames (51 MB), TRES-1190603072712.FITS → TRES-1190603111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-190603.log (0bda31681bef...), FinalLightCurve_TrES-1 b_2019-06-03.png (c1016f482735...), FinalLightCurve_TrES-1 b_2019-06-03.csv (b6de034017ef...)

Compute resources

Wall time 265.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:32Z.